Chapter 6

Simplifications

A complex arithmetical expression can be converted into a simple expression by **simplification**.

6.1 VBODMAS' Rule

To simplify arithmetic expressions, which involve various operations like brackets, multiplication, addition, etc; a particular sequence of the operations has to be followed. The operations have to be carried out in the order, in which they appear in the word **VBODMAS**, where different letters of the word stand for following operations.

$$\left[\left\{ \left(- \right) \right\} \right]$$

Order of removing brackets

1. Small brackets (Circular brackets) '( )'
2. Middle brackets (Curly brackets)'{ }'
3. Square brackets (Big brackets) '[ ]'

The letters in ODMAS are the operations: **O** =Of; **D** = Division; **M** = Multiplication; **A** = Addition; **S** = Subtraction
Order of above mentioned operations is same as the order of letters in the 'VBODMAS' from left to right as
Clearly, the order will be as follows

First **Vinculum bracket** is solved [Remember -5-10=-15, but $-\overline{5-10} = -(-5) = 5$]

Second **Brackets** are to be solved in order given above, [first,then second, then third]

Third Operation of **'OF'** is done

Fourth Operation of **division** is performed

Fifth Operation of **multiplication** is performed

Sixth Operation of **addition** is performed

Seventh Operation of **subtraction** is performed

Note

Absolute value of a real number
If m is a real number, then its absolute value is defined as

$$|m| = \begin{cases} m & \text{if } m \geq 0 \\ -m & \text{if } m < 0 \end{cases}$$

For example: $|3| = 3$ and $|-3| = -(-3) = 3$.
Ex. 1 Simplify $4 - [6 - \{12 - (10 - \overline{8-6})\}]$

Sol. Given expression $= 4 - [6 - \{12 - (10 - \overline{8-6})\}]$

$= 4 - [6 - \{12 - (10 - 8 + 6)\}]$ Remove vinculum

$= 4 - [6 - \{12 - (10 - 2)\}]$

$= 4 - [6 - \{12 - 8\}]$ Remove ()

$= 4 - [6 - 4]$ Remove {}

$= 4 - 2 = 2$ Remove []

Ex.2 Simplify $\left(9.6 \times 3.6 \div 7.2 + 10.8 \text{ of } \frac{1}{18} - \frac{1}{10}\right)$

Sol. Given expression $= 9.6 \times 3.6 \div 7.2 + 10.8 \text{ of } \dfrac{1}{18} - \dfrac{1}{10}$

$= 9.6 \times 3.6 \div 7.2 + 0.6 - \dfrac{1}{10}$ $\left[\text{Remove 'of'- } \quad 10.8 \text{ of } \dfrac{1}{18} = 10.8 \times \dfrac{1}{18}\right]$

$= 9.6 \times \dfrac{3.6}{7.2} + 0.6 - 0.1$ Remove ÷

$= 9.6 \times \dfrac{1}{2} + 0.6 - 0.1$

$= 4.8 + 0.6 - 0.1$ Remove ×

$= 5.4 - 0.1$ Remove +

$= 5.3$ Remove -

Ex.3 Simplify $18800 \div 940 \div 10$

Sol. Given expression $= 18800 \div 940 \div 10$

$= \dfrac{18800}{940} \div 10$

$= 20 \div 10 = 2$

Ex.4 Solve $\quad 4\dfrac{1}{6} + 5\dfrac{1}{6} + 8\dfrac{1}{6}.$

Sol. Given expression $= 4\dfrac{1}{6} + 5\dfrac{1}{6} + 8\dfrac{1}{6}$

$= (4+5+8) + \dfrac{1}{6} + \dfrac{1}{6} + \dfrac{1}{6}$

$= 17 + \dfrac{3}{6} = 17\dfrac{1}{2}$

Ex.5 Simplify $n - [n - (m+n) - \{n - (n - \overline{m-n})\} + 2m]$

Sol. Given expression $= n - [n - (m+n) - \{n - (n - \overline{m-n})\} + 2m]$

$= n - [n - (m+n) - \{n - (n - m + n)\} + 2m]$

$= n - [n - (m+n) - \{n - (2n - m)\} + 2m]$

$= n - [n - m - n - \{n - 2n + m\} + 2m]$

$= n - [-m - \{-n + m\} + 2m]$

$= n - [-m + n - m + 2m]$

$= n - [n]$

$= n - n = 0$

Ex.6 Simplify $\dfrac{\frac{7}{2} \div \frac{5}{2} \times \frac{3}{2}}{\frac{7}{2} \div \frac{5}{2} of \frac{3}{2}} \div \dfrac{15}{14}$

Sol. Given expression $= \dfrac{\frac{7}{2} \div \frac{5}{2} \times \frac{3}{2}}{\frac{7}{2} \div \frac{5}{2} of \frac{3}{2}} \div \dfrac{15}{14}$

$$= \frac{\frac{7}{2} \div \frac{5}{2} \times \frac{3}{2}}{\frac{7}{2} \div \frac{15}{4}} \div \frac{15}{14}$$

$$= \frac{\frac{7}{2} \times \frac{2}{5} \times \frac{3}{2}}{\frac{7}{2} \times \frac{4}{15}} \div \frac{15}{14} = \frac{\frac{21}{10}}{\frac{14}{15}} \div \frac{15}{14}$$

$$= \frac{21}{10} \times \frac{15}{14} \times \frac{14}{15} = \frac{21}{10} = 2\frac{1}{10}$$

Basic Formulae

Following formulae are useful in various operations of simplification

(i) $(a+b)^2 = a^2 + 2ab + b^2$

(ii) $(a-b)^2 = a^2 - 2ab + b^2$

(iii) $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$

(iv) $(a+b)^2 - (a-b)^2 = 4ab$

(v) $a^2 - b^2 = (a+b)(a-b)$

(vi) $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$

(vii) $(a-b)^3 = a^3 - b^3 - 3ab(a-b)$

(viii) $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$

(ix) $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

(x)

$$\frac{a^3 + b^3 + c^3 - 3abc}{(a^2 + b^2 + c^2 - ab - bc - ca)} = (a+b+c)$$

$$OR$$

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$= (a+b+c)\frac{1}{2}(2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca)$$

$$= \frac{(a+b+c)}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$$

If $a+b+c=0$ then $a^3+b^3+c^3=3abc$ or $a^3+b^3+c^3-3abc=0$

(xi) $(a+b+c)^2=(a^2+b^2+c^2)+2(ab+bc+ca)$

(xii) $\left(a+\frac{1}{a}\right)^2=\left(a^2+\frac{1}{a^2}\right)+2=\left(a-\frac{1}{a}\right)^2+4$

(xiii) $\left(a-\frac{1}{a}\right)^2=\left(a^2+\frac{1}{a^2}\right)-2=\left(a+\frac{1}{a}\right)^2-4$

(xiv) $\left(a+\frac{1}{a}\right)^3=\left(a^3+\frac{1}{a^3}\right)+3\left(a+\frac{1}{a}\right)$

(xv) $\left(a-\frac{1}{a}\right)^3=\left(a^3-\frac{1}{a^3}\right)-3\left(a-\frac{1}{a}\right)$

Ex.7 Solve $\frac{(5.9)^3+(1.8)^3+(4.8)^3-3\times5.9\times1.8\times4.8}{(5.9)^2+(1.8)^2+(4.8)^2-5.9\times1.8-1.8\times4.8-4.8\times5.9}$

Sol. We know that

$$\frac{a^3+b^3+c^3-3abc}{(a^2+b^2+c^2-ab-bc-ca)}=(a+b+c)$$

Here, $a=5.9$, $b=1.8$, $c=4.8$

$$\therefore \quad \frac{(5.9)^3+(1.8)^3+(4.8)^3-3\times5.9\times1.8\times4.8}{(5.9)^2+(1.8)^2+(4.8)^2-5.9\times1.8-1.8\times4.8-4.8\times5.9}$$
$$=(5.9+1.8+4.8)=12.5$$

Ex.8 Solve $\frac{(9.8)^3-(6.8)^3}{9.8^2+9.8\times6.8+6.8^2}$

Sol. We know that

$$a^3-b^3=(a-b)(a^2+ab+b^2)$$

$$\therefore \quad \frac{a^3-b^3}{(a^2+ab+b^2)}=(a-b)$$

Here, $a=9.8, b=6.8$

$$\therefore \quad \frac{(9.8)^3-(6.8)^3}{9.8^2+9.8\times6.8+6.8^2}=(9.8-6.8)=3$$

Ex.9 Solve $\dfrac{(835+378)^2+(835-378)^2}{835\times 835+378\times 378}$

Sol. We know that

$$\frac{(a+b)^2+(a-b)^2}{a^2+b^2}=\frac{2(a^2+b^2)}{a^2+b^2}=2$$

$$\therefore \quad \frac{(835+378)^2+(835-378)^2}{835\times 835+378\times 378}=2$$

Ex.10 If $x=3+\sqrt{8}$, then find the value of $x^2+\dfrac{1}{x^2}$

Sol. Given that $x=3+\sqrt{8}$

On squaring both sides we get

$$x^2=(3+\sqrt{8})^2=9+8+6\sqrt{8} \qquad [\because \quad (a+b)^2=a^2b+^2+2ab]$$

$$\frac{1}{x^2}=17+6\sqrt{8}=17+12\sqrt{2}$$

Now, $$\frac{1}{x^2}=\frac{1}{17+12\sqrt{2}}\times\frac{17-12\sqrt{2}}{17-12\sqrt{2}} \qquad \text{[rationalising the denominator]}$$

$$\Longrightarrow \qquad \frac{1}{x^2}=\frac{17-12\sqrt{2}}{289-288}=17-12\sqrt{2}$$

$$\therefore \qquad x^2+\frac{1}{x^2}=(17+12\sqrt{2})(17-12\sqrt{2})$$

$$x^2+\frac{1}{x^2}=34$$

Ex.11 If $a+\dfrac{1}{a}=4\sqrt{2}$, then find the value of (a^6+a^{-6})

Sol. Given $a+\dfrac{1}{a}=4\sqrt{2}$

On squaring both sides, we get,

$$\left(a+\frac{1}{a}\right)^2=4\sqrt{2}^2$$

$$\Longrightarrow \quad 4\sqrt{2}^2=a^2+\frac{1}{a^2}+2 \qquad [\because (a+b)^2=a^2+b^2+2ab]$$

$$\Longrightarrow a^2+\frac{1}{a^2}=32-2=30$$

Now, taking cube on both sides, we get,

$$\left(a^2+\frac{1}{a^2}\right)^3 = 30^3$$

$$\Longrightarrow \quad a^6+\frac{1}{a^6}+3a^2.\frac{1}{a^2}\left(a^2+\frac{1}{a^2}\right)=27000 \qquad [\because \quad (a+b)^3 = a^3+b^3+3ab(a+b)]$$

$$\Longrightarrow \quad a^6+\frac{1}{a^6}+3\times 30 = 27000$$

$$\Longrightarrow \quad a^6+\frac{1}{a^6} = 27000-90 = 26910$$

$$\therefore \quad a^6+a^{-6} = 26910$$